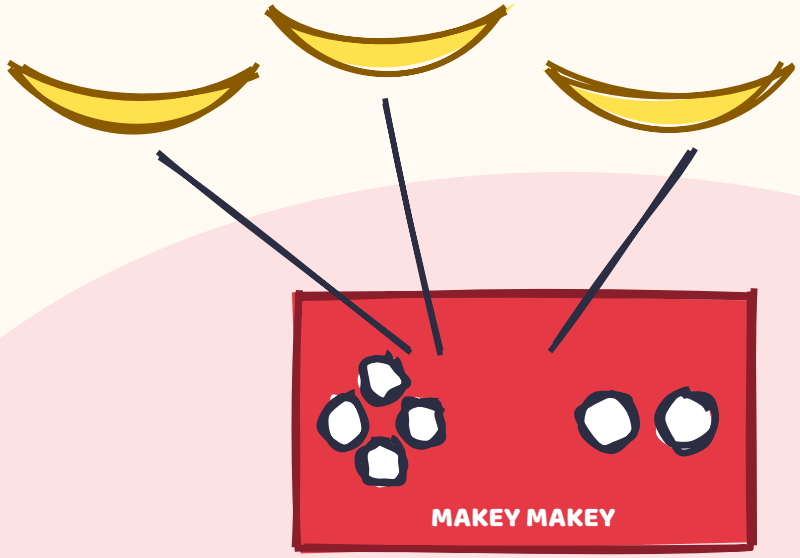
Little Explorers · Class 13

Makey Makey
Fruit Piano

Play music on bananas!

Conductors | Makey Makey | Class band



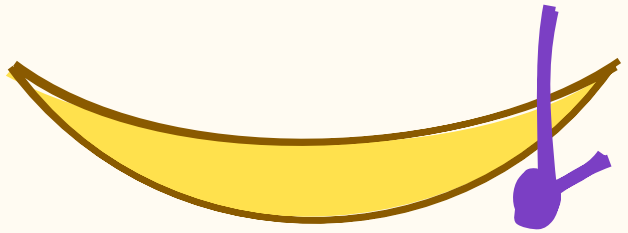
Did you know?
Makey Makey was invented by two students at MIT who wanted everyone to be an inventor!

Let's wonder...

Can a banana be a piano key?

A fruity mystery

Computers have keyboards with buttons. But what if **anything** could be a button: a banana, an apple, even your friend? Let's find out!



No way!

Maybe?

Let's try!

Today: we turn fruit into a **piano** with a Makey Makey!

LEARN

Learning objectives

Today I can...

1	**Conductors:** sort things into **conductors** and **insulators**	**Proof:** our sorting sheet has both groups
2	**Fruit keys:** turn fruit into **computer keys** with a Makey Makey	**Proof:** I play a note by touching a banana
3	**Play together:** play a simple tune as a **class band**	**Proof:** our band played a tune together

Objective 1 · Conductors · video

Video time!



Conductors and Insulators

Little School · about 3 minutes · youtu.be/iZ6IQ51u6T8

Watch for...

1. What is a **conductor**?
2. What is an **insulator**?
3. Is **water** a conductor or an insulator?

Scan to watch

After the video: Point to something in the room you think is a conductor. Your partner points to an insulator.

Objective 1 · Conductors

Some things let electricity through

The idea

Conductors let a tiny bit of electricity flow through them: metal, foil, fruit, water, and **people**!

Insulators stop it: plastic, wood, paper, rubber, cloth.

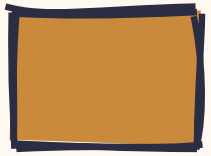
Safe fun

Makey Makey uses a very, very tiny current. It is completely safe to touch!

Example: a water slide



conductor



insulator

Water zooms down a slide (a **conductor**). Put a wooden block in the way and the water stops (an **insulator**). Fruit is juicy, so electricity can slide through it!

Real world

Touch is everywhere



Touch screens
Feel the tiny electricity in your finger



Electronic music
Musicians invent new instruments



Touch lamps
Tap the metal base to switch on



Helping people
Special buttons for people who cannot press hard

Why does a touch screen not work with gloves on?

Clue
Gloves are insulators!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Makey Makey rules
2. Conductor or insulator?
3. Meet the Makey Makey
4. Build the fruit piano
5. Human chain
6. Our class band
7. Fruit piano challenges

Stay safe

Makey Makey rules

Safe Makey Makey uses tiny, safe electricity.

Never touch wall sockets or plugs. They are NOT safe.

Wash hands after touching fruit.

Eat later? Only fruit that was not clipped. Clipped fruit goes in the compost.



Remember

Tiny current from a USB cable is safe. Big current from the wall is dangerous!

Sort it!

Conductor or insulator?

Thing	I think…	Did it play a note?
Banana	conductor / insulator?	_____
Spoon	conductor / insulator?	_____
Pencil (wood)	conductor / insulator?	_____
Foil	conductor / insulator?	_____
Plastic cup	conductor / insulator?	_____

Talk about it

Were you surprised by any results? Try a **pencil drawing** on paper: the grey graphite conducts!

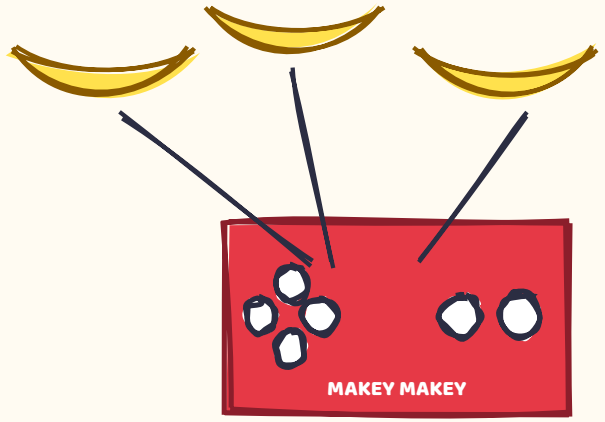
Hold earth

Hold the earth wire with one hand and touch the thing with the other.

MAKE

Objective 2 · Fruit keys · how it works

Meet the Makey Makey



Clip and touch

Clip a fruit to a key (like the **space** or **arrow** keys). Hold the **earth** wire. Touch the fruit: the computer thinks you pressed that key!

Earth wire

No one holding earth? No note! Try it.

You are part of the circuit!

The tiny current goes through the fruit, **through you**, and back through the earth wire.

MAKE

Objective 2 · Build step 1 of 2

Build the fruit piano

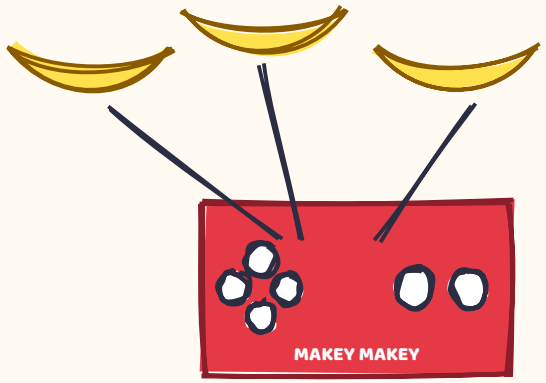
a Plug the Makey Makey into the laptop with the USB cable.

b Clip one wire to each **fruit** and to an **arrow key** or **space**.

c Clip one wire to **EARTH**. One child holds the end.

d Open the **Makey Makey piano** app.

e Hold earth and tap a banana: **do!**



Check: Every fruit plays a different note.

MAKE

Class game

Human chain

How to play

1. Hold the earth wire with one hand.
2. Hold hands with friends to make a long chain.
3. The last child touches a banana.
4. Does the note still play? **Yes!**
5. One child lets go… what happens?

Then discuss

Why did the note stop when one child let go?



Big idea

People are conductors too. A broken chain is a broken circuit!

MAKE

Objective 3 · Play together

Our class band

Song	Fruit keys to tap
Twinkle Twinkle	1 1 5 5 6 6 5
Happy Birthday (start)	5 5 6 5 1 7
Our own tune	_ _ _ _ _ _ _

Take turns

Each child looks after **one** fruit. The conductor (a child!) points to who plays next.

Numbers

Stick a number sticker on each fruit: 1 to 7.

MAKE

Level up!

Fruit piano challenges

Choose your challenge



Weird keys: try a potato, a cup of water or a pencil drawing.



Fruit game: use the arrow keys to play a game with fruit.



High five key: make a high five play a note!



Drum kit: turn the app to drums and play a beat.

Think like an engineer

What other things could you turn into keys?

Boss level

Play a whole song with no mistakes!

Objective check

Can I really do it? Prove it.

01	Name one conductor and one insulator.	○ ○ ○
02	Show how to play a note on a banana.	○ ○ ○
03	Did our class band play a tune together?	○ ○ ○

Rate yourself
Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Make a big class poster: two columns, conductors and insulators.

Show what you know

What did we learn?

- [x] Conductors let electricity through: fruit, metal, people
- [x] Insulators stop it: plastic, wood, rubber
- [x] Makey Makey turns anything conductive into a key
- [x] We are part of the circuit!

At home
Find 3 conductors and 3 insulators at home.



Exit ticket: is a plastic spoon a conductor or an insulator?

Next class: make an Animated Story App!